Fast and Latent Low-Rank Subspace Clustering for Hyperspectral Band Selection

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***Abstract*— This article presents a fast and latent low-rank subspace clustering (FLLRSC) method to select hyperspectral bands. The FLLRSC assumes that all the bands are sampled from a union of latent low-rank independent subspaces and formulates the self-representation property of all bands into a latent low-rank representation (LLRR) model. The assumption ensures sufficient sampling bands in representing low-rank subspaces of all bands and improves robustness to noise. The FLLRSC first implements the Hadamard random projections to reduce spatial dimensionality and lower the computational cost. It then adopts the inexact augmented Lagrange multiplier algorithm to optimize the LLRR program and estimates sparse coefficients of all the projected bands. After that, it employs a correntropy metric to measure the similarity between pairwise bands and constructs an affinity matrix based on sparse representation. The correntropy metric could better describe the nonlinear characteristics of hyperspectral bands and enhance the block-diagonal structure of the similarity matrix for correctly clustering all subspaces. The FLLRSC conducts spectral clustering on the connected graph denoted by the affinity matrix. The bands that are closest to their separate cluster centroids form the final band subset. Experimental results on three widely used hyperspectral data sets show that the FLLRSC performs better than the classical low-rank representation methods with higher classification accuracy at a low computational cost.**

***Index Terms*— Band selection, correntropy measure, hyperspectral imagery (HSI), latent low-rank subspace clustering, remote sensing.**

I. Introduction

HYPERSPECTRAL sensor collects both spectral and spatial information of ground objects on the earth surface using hundreds of bands [1]–[3]. The obtained hyperspectral imagery (HSI) can be used to identify different materials with subtle spectral divergences, which benefits many realistic applications, e.g., land cover mapping [4], coastal land monitoring [5], precision agriculture [6], and mine exploration [7]. However, the high dimensionality of HSI bands and strong inter-band correlations bring about spectral information redundancy and high computation burden [8], [9]. Moreover, the "Hughes" problem occurs when an improbably large number of training samples are demanded to ensure high classification accuracy [10], [11]. Therefore, dimensionality reduction is an option to conquer the above problems for fine classification.

In contrast with feature extraction, band selection is to select a proper band subset from the original set and well preserves the spectral meaning of dimensionality-reduced data [12]. Many classical band selection methods have been presented in the literature, such as ranking-based [13], clustering-based [14], searching-based [15], embedding learning-based [16], and sparsity-based methods. The ranking-based methods, e.g., maximum-variance principal component analysis (MVPCA) [13] and the manifold ranking-based band selection algorithm [17], quantify the importance of each band and select the top-ranked bands. The selected bands highly depend on the defined band prioritization criterion [18], and the selected bands may have high information redundancy since the ranking-based methods do not consider the inter-band correlation. The clustering-based methods, e.g., the fast density-peak-based clustering (FDPC) algorithm [19] and the optimum clustering framework (OCF) algorithm [20], use a clustering technique to select bands and can avoid selecting highly correlated bands. Unfortunately, the selected bands do not necessarily have maximum information, and the random initialization in clustering brings about high uncertainties for the selected bands [12]. The searching-based methods, e.g., the fire fly algorithm [21] and the multigraph determinant point process algorithm [22], transform the band selection problem into an optimization problem to select proper bands. However, the nonlinear optimization problems always have a high computational complexity. The embedding learning-based methods, e.g., recursive support vector machines (SVMs) [16] and sparse multinomial logistic regression algorithms [23], formulate band selection into an optimization program of specific application models (i.e., classifier and target detection), where band subset and application models are simultaneously estimated by optimizing the defined optimization program. However, the prior knowledge of training samples may be unavailable in some scenarios [24], which limits the applications of embedding learning-based methods.

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In recent years, the popularity of sparsity theory brings a new perspective for hyperspectral band selection [12], [25], [26]. The sparsity theory states that each band can be sparsely represented by several other bands, and sparse coefficients reveal certain underlying structures within the HSI bands and hence benefit selecting proper bands. Typical methods include sparse nonnegative matrix factorization (SNMF)-based algorithms [27] and the sparse representation-based algorithms [28]. The SNMF-based algorithms decompose the HSI data matrix into the product of an unknown basis matrix and an unknown coefficient matrix with both sparsity and nonnegativity constraints, and the band set is selected by clustering sparse coefficients [29]. The selected bands from the SNMF lack clear physical or geometric meanings [30]. Compared against the SNMF, the sparsity-based algorithms learn the dictionary in advance and then select the informative bands using the estimated sparse coefficients, e.g., multitask sparse learning algorithm [15] and collaborative sparse algorithm [31]. Particularly, when the dictionary is set to be the HSI band matrix itself, all the bands show low-rank and sparse self-representation property [25]. Accordingly, all the HSI bands are regarded to be sampled from several low-rank independent subspaces, and the subspaces can be demonstrated via the block diagonal structure of estimated sparse coefficients. The subspace clustering-based algorithms select representative bands by spectral clustering on the sparse coefficients, and typical examples include improved sparse subspace clustering (ISSC) [25], nonnegative low-rank representation (NNLRR) [32], dissimilarity weighted sparse self-representation [26], and fast and robust self-representation [12]. However, most current subspace clustering algorithms for band selection suffer two problems: sufficient band samples (i.e., bands) are required to ensure well representation of underlying subspaces, and the subspace clustering is suitable to noiseless data to achieve a robust segmentation of subspaces. Unfortunately, HSI bands usually contain noise, which degrades the performance of clustering.

In this article, we would like to handle the above problems by presenting a fast and latent low-rank subspace clustering (FLLRSC) method for hyperspectral band selection. The FLLRSC introduces unobserved HSI bands to improve the traditional low-rank representation (LRR) model with the assumption that all the bands are sampled from a series of latent low-rank subspaces spanned by both observed and unobserved HSI bands. The unobserved HSI bands characterize hidden features during hyperspectral imaging procedure, and they guarantee sufficiently sampling bands of the self-representation dictionary in representing low-rank subspaces and improve the robustness of subspace clustering to noise [33]. Finding latent LRRs (LLRRs) of all bands can be transformed into a nuclear norm minimization problem, which can be solved by the inexact augmented Lagrange multiplier (IALM) algorithm [34]. Considering that a large number of HSI pixels might bring about extremely high computational cost, the Hadamard random projection (HRP) is used to pre-reduce the spatial size of HSI bands. Meanwhile, the correntropy measure [35] is adopted to describe the nonlinear characteristics of HSI bands. In detail, the FLLRSC constructs a correntropy similarity matrix with the sparse coefficient matrix of all the observed bands, and then performs spectral clustering on the graph of the similarity matrix and selects the bands closest to their cluster centroids.

Compared against previous subspace clustering-based band selection methods, our article favors the following three contributions.

1) Based on our knowledge, this is the first time for the LLRR to be introduced into modeling the problem of hyperspectral band selection. The FLLRSC considers hidden HSI effects to construct the LLRR model of all the observed bands, which helps to resolve the problems of both insufficient samplings of the self-representation dictionary and the sensitivity to noise in current subspace clustering-based methods.
2) The FLLRSC considers the distinctive characteristics of both nonlinearity and high spatial-dimensionality in HSI bands, and utilizes the correntropy measure to ensure a block-diagonal structure of the similarity matrix for more accurate clustering of all subspaces. The HRPs in FLLRSC can greatly reduce computational cost in band selection whereas do not degrade its performance.
3) The experimental results demonstrate that the FLLRSC can achieve higher overall classification accuracies (OCAs) than using all bands when a moderate size of band subset size is selected [e.g., 20 on the Pavia University (PaviaU)].

The rest of this article is arranged as follows. Section II briefly reviews the LRR model for spectral clustering. Section III describes the FLLRSC for hyperspectral band selection. Section IV presents experimental results on three hyperspectral data sets. Section V draws conclusions of this article.

II. LRR for Spectral Clustering

Let $\mathbf{X} = \{\mathbf{x}_i\}_{i=1}^{N} \in \mathbf{R}^{M\times N}$ be a collection of high-dimensional data samples, where M is the dimensionality of feature space and N is the number of samples. The subspace membership of the samples is determined by the row space of $\mathbf{X}$. The LRR assumes that all the samples are drawn from a union of independent subspaces [36], and aims to find the lowest-rank representation among all the data samples that can represent them as linear combinations of the basis in a given dictionary. The formulation of LRR can be written as

$$\min_{\mathbf{Z},\mathbf{E}} \operatorname{rank}(\mathbf{Z}) + \mu\|\mathbf{E}\|_l, \quad \text{s.t.,} \ \mathbf{X} = \mathbf{A}\mathbf{Z} + \mathbf{E} \tag{1}$$

where $\mathbf{A}$ denotes the dictionary that linearly spans the data space, $\mathbf{Z}$ is the coefficient matrix, and $\mu > 0$ is the regularization parameter. Since $\operatorname{rank}(\mathbf{AZ}) \le \operatorname{rank}(\mathbf{Z})$, the term $\operatorname{rank}(\mathbf{Z})$ guarantees the lowest-rank property of data $\mathbf{X}$ with respect to dictionary $\mathbf{A}$. The $\|\mathbf{E}\|_l$ is the regularization term that characterizes random noise and sample-specific corruptions and outliers. The nuclear norm $\|\mathbf{E}\|_{2,1}$ is a good surrogate to replace the rank function, and also a good relaxation to denote the sample-specific corruptions and outliers in the data samples. When the data matrix itself $\mathbf{X}$ is used as the dictionary (i.e., $\mathbf{A} = \mathbf{X}$), LRR is transformed into the convex

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optimization problem of subspace clustering [32]

$$\min_{\mathbf{Z},\mathbf{E}} \|\mathbf{Z}\|_* + \mu\|\mathbf{E}\|_{2,1}, \quad \text{s.t.,}\ \mathbf{X} = \mathbf{X}\mathbf{Z} + \mathbf{E} \tag{2}$$

which can be solved by various methods, such as the augmented Lagrange multiplier method. The estimated coefficient matrix $\mathbf{Z}$ can help to cluster all data points into its belonged subspace.

III. FLLRSC for Band Selection

Consider a high-dimensional HSI data as a matrix $\mathbf{X} = \{\mathbf{x}_i\}_{i=1}^N \in \mathbb{R}^{M\times N}$, where each band of $\mathbf{X}$ is sampled from a union of k subspaces, M is the number of pixels in the image scene, and N is the number of bands with $N \ll M$. Selecting a band subset $\mathbf{M} = \mathbf{X}(:, \Omega) = \{\mathbf{m}_i\}_{i=1}^k \in \mathbb{R}^{M\times k}$ can be regarded as finding the k representative bands $\mathbf{M}$ from the original band set, where $\Omega \in \{1, 2, \cdots, N$ and $|\Omega| = k$.

A. Spatial Dimensionality Reduction Using HRPs

Hyperspectral remote sensing images have millions of pixels, which brings about high computational burden for band selection. Random projections originate from the famous Johnson–Lindenstrauss lemma and have been proven to be an efficient method for dimensionality reduction [8]. Compared against the regular Gaussian random projections, the HRPs have lower computational costs and better performance in preserving the main information of original data [37]. Therefore, we adopt HRPs to sketch the HSI data matrix $\mathbf{X}$ from rows and reduce the spatial dimensionality of $\mathbf{X}$. The Hadamard random matrix-based dimensionality reduction is defined as

$$\mathbf{Y}_O = \mathbf{\Phi}^{\mathrm{T}}\mathbf{X} = \left(\sqrt{\frac{K}{M}}\mathbf{D}\mathbf{H}_M\mathbf{P}\right)^{\mathrm{T}}\mathbf{X} \tag{3}$$

where $\mathbf{Y}_O \in \mathbb{R}^{K\times N}$ is the projected matrix, and $\mathbf{\Phi} = \sqrt{(K/M)}\mathbf{D}\mathbf{H}_M\mathbf{P} \in \mathbb{R}^{M\times K}$ is the HRPs matrix. $\mathbf{D} \in \mathbb{R}^{M\times M}$ is a diagonal matrix with diagonal entries sampled uniformly from $\{-1, 1\}$. $\mathbf{H}_M \in \mathbb{R}^{M\times M}$ is the Hadamard matrix defined recursively for any M, that is an integer power of 2 as

$$\mathbf{H}_M = \begin{bmatrix} \mathbf{H}_{2/M} & \mathbf{H}_{2/M} \\ \mathbf{H}_{2/M} & -\mathbf{H}_{2/M} \end{bmatrix}.$$

$\mathbf{P} \in \mathbb{R}^{M\times K}$ is a uniform sampling matrix that randomly samples K columns of $\mathbf{D}\mathbf{H}_M$, where each column of $\mathbf{P}$ is randomly selected with replacement from the $M \times M$ identity matrix $\mathbf{I}_M$. The spatial dimensionality of original HSI data $\mathbf{X}$ is reduced into K, and its computational complexity is reduced to $O(MN\log(K))$.

B. Modeling the Latent Low-Rank Subspace With Projected HSI Band Matrix

After HRPs, the FLLRSC introduces some unobserved hidden noisy HSI bands to concatenate with the self-representation dictionary of LRR and formulates the LLRR model of all the projected bands. The unobserved hidden bands are assumed to be sampled from the same collection of low-rank subspaces with the original HSI data (i.e., observed bands). Taking the concatenation (along columns) of original and unobserved projected HSI bands as the dictionary, the original projected bands can be expressed as a sparse linear combination of other bands within the same low-rank subspace, and the projected bands can be formulated as the following problem:

$$\begin{aligned} \min_{\mathbf{Z}} \|\mathbf{Z}\|_*, \quad \text{s.t.,}\ \mathbf{Y}_O &= [\mathbf{Y}_O, \mathbf{Y}_H]\mathbf{Z} + \mathbf{E} \\ \operatorname{rank}(\mathbf{Y}_O) &= \mathbf{rank}(\mathbf{Y}_H) = r \end{aligned} \tag{4}$$

where $\mathbf{Y}_O$ is the original and observed projected band matrices and $\mathbf{Y}_H$ is the unobserved projected band matrix. $\mathbf{Z}$ is the coefficient matrix with the row-wise partition $\mathbf{Z} = [\mathbf{Z}_O; \mathbf{Z}_H]$, and $\mathbf{Z}_O$ and $\mathbf{Z}_H$ correspond to the coefficient matrices of $\mathbf{Y}_O$ and $\mathbf{Y}_H$, respectively. The term $\|\mathbf{Z}\|_*$ is to restrict the low-rank property of subspaces. The concatenated dictionary $[\mathbf{Y}_O, \mathbf{Y}_H]$ ensures the sufficient bands in all subspaces and could well represent its low-rank structure. Moreover, the unobserved projected band matrix $\mathbf{Y}_H$ characterizes the noise data in the HSI imaging and improves the robustness of the model. But the above problem is ill-posed because the coefficient matrix is determined by both observed projected band matrix $\mathbf{Y}_O$ and unknown hidden projected band matrix $\mathbf{Y}_H$.

In [36], it has been proven that the minimizer to problem (3) with noiseless data (i.e., $\mathbf{E} = 0$) is unique and has the following closed solution $\mathbf{Z}_O = \mathbf{V}_O\mathbf{V}_O^{\mathrm{T}}$ and $\mathbf{Z}_H = \mathbf{V}_H\mathbf{V}_O^{\mathrm{T}}$, where $\mathbf{V}_O$ and $\mathbf{V}_H$ are estimated from the singular value decomposition (SVD) of $[\mathbf{Y}_O, \mathbf{Y}_H] = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^{\mathrm{T}}$ and then partition $\mathbf{V}$ such that $\mathbf{Y}_O = \mathbf{U}\mathbf{\Sigma}\mathbf{V}_O^{\mathrm{T}}$ and $\mathbf{Y}_H = \mathbf{U}\mathbf{\Sigma}\mathbf{V}_H^{\mathrm{T}}$. Equation (4) is transformed into

$$\begin{aligned} \mathbf{Y}_O &= [\mathbf{Y}_O, \mathbf{Y}_H]\,\mathbf{Z} + \mathbf{E} = \mathbf{Y}_O\mathbf{Z}_O + \mathbf{Y}_H\mathbf{Z}_H + \mathbf{E} \\ &= \mathbf{Y}_O\mathbf{Z}_O + \mathbf{U}\Sigma\mathbf{V}_H^{\mathrm{T}}\mathbf{V}_H\mathbf{\Sigma}^{-1}\mathbf{U}^{-1}\mathbf{Y}_O + \mathbf{E} \\ &= \mathbf{Y}_O\mathbf{Z}_O + \mathbf{L}_H\mathbf{Y}_O + \mathbf{E} \end{aligned} \tag{5}$$

where $\mathbf{L}_H = \mathbf{U}\Sigma\mathbf{V}_H^{\mathrm{T}}\mathbf{V}_H\mathbf{\Sigma}^{-1}\mathbf{U}^{-1}$. From the above, $\mathbf{Y}_O$ and $\mathbf{Y}_H$ are assumed to be sampled from the same collections of low-rank subspaces, and hence the coefficient matrices $\mathbf{Z}_O$ and $\mathbf{L}_H$ also have the same low rank with $\operatorname{rank}(\mathbf{Z}_O) = \mathbf{rank}(\mathbf{L}_H) \le r$. Furthermore, following the sample-specific corruptions and outliers in program (2), the problem (3) can be formulated into the LLRR model in:

$$\begin{aligned} &\min_{\mathbf{Z}_O\mathbf{L}_H} \|\mathbf{Z}_O\|_* + \|\mathbf{L}_H\|_* + \lambda\|\mathbf{E}\|_{2,1} \\ &\text{s.t.,}\ \mathbf{Y}_O = \mathbf{Y}_O\mathbf{Z}_O + \mathbf{L}_H\mathbf{Y}_O + \mathbf{E}. \end{aligned} \tag{6}$$

$\mathbf{Z}_O$ has the diagonal-block structure and can reveal the clustering structure of bands from the same subspace, and the $\mathbf{L}_H$ demonstrates hidden effects within the collected HSI bands.

C. Solving the LLRR Program

The objective function (6) is convex and can be solved by the IALM [34]. First, two auxiliary variables $\mathbf{J} = \mathbf{Z}_O$ and $\mathbf{S} = \mathbf{L}_H$ are introduced, and (6) becomes

$$\begin{aligned} &\min_{\mathbf{Z}_O\mathbf{L}_H,\mathbf{E}} \|\mathbf{J}\|_* + \|\mathbf{S}\|_* + \lambda\|\mathbf{E}\|_{2,1}, \\ &\text{s.t.}\ \mathbf{Y}_O = \mathbf{Y}_O\mathbf{Z}_O + \mathbf{L}_H\mathbf{Y}_O + \mathbf{E}, \mathbf{Z}_O = \mathbf{J}, \quad \mathbf{L}_H = \mathbf{S}. \end{aligned} \tag{7}$$

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The augmented Lagrange function of (7) can be written as

$$\begin{aligned}\mathcal{L}(&\mathbf{Z}_O, \mathbf{L}_H, \mathbf{E}, \mathbf{J}, \mathbf{S}, \mathbf{\Lambda}, \mathbf{\Delta}, \beta)\\ &= \operatorname{argmin}\|\mathbf{J}\|_* + \|\mathbf{S}\|_* + \lambda\|\mathbf{E}\|_{2,1}\\ &\quad + \operatorname{tr}\big(\mathbf{\Delta}_1^{\mathrm{T}}\big(\mathbf{Y}_O - \mathbf{Y}_O\mathbf{Z}_O - \mathbf{L}_H\mathbf{Y}_O - \mathbf{E}\big)\big) + \operatorname{tr}\big(\mathbf{\Delta}_2^{\mathrm{T}}(\mathbf{Z}_O - \mathbf{J})\big)\\ &\quad + \operatorname{tr}\big(\mathbf{\Delta}_3^{\mathrm{T}}(\mathbf{L}_H - \mathbf{S})\big) + \frac{\beta}{2}\big(\|\mathbf{Y}_O - \mathbf{Y}_O\mathbf{Z}_O - \mathbf{L}_H\mathbf{Y}_O - \mathbf{E}\|_F^2\\ &\quad + \|\mathbf{Z}_O - \mathbf{J}\|_F^2 + \|\mathbf{L}_H - \mathbf{S}\|_F^2\big) \qquad (8)\end{aligned}$$

where $\mathbf{\Delta}_1 \in \mathbb{R}^{K\times K}$, $\mathbf{\Delta}_2 \in \mathbb{R}^{K\times K}$, and $\mathbf{\Delta}_3 \in \mathbb{R}^{K\times K}$ are Lagrange multipliers and $\beta > 0$ is the selected penalty parameter.

Then the IALM optimizes the seven variables with iterative procedures and updates each variable at iteration $t+1$ using the following schemes. When fixing variables $\mathbf{Z}_O^{(t)}$ and $\mathbf{\Delta}_2^{(t)}$, the variable $\mathbf{J}^{(t+1)}$ can be updated as

$$\mathbf{J}^{(t+1)} = \underset{\mathbf{J}}{\operatorname{argmin}}\ \|\mathbf{J}\|_* + \frac{1}{2}\left\|\mathbf{J} - \left(\mathbf{Z}_O^{(t)} + \frac{\mathbf{\Delta}_2^{(t)}}{\beta^{(t)}}\right)\right\|_F^2. \qquad (9)$$

When fixing $\mathbf{L}_H$ and $\mathbf{\Delta}_3$, the $\mathbf{S}^{(t+1)}$ can be updated as

$$\mathbf{S}^{(t+1)} = \underset{\mathbf{s}}{\operatorname{argmin}}\ \|\mathbf{S}\|_* + \frac{1}{2}\left\|\mathbf{S} - (\mathbf{L}_H^{(t)} + \frac{\mathbf{\Delta}_3^{(t)}}{\beta^{(t)}})\right\|_F^2. \qquad (10)$$

When fixing the variables $\mathbf{L}_H^{(t)}$, $\mathbf{E}^{(t)}$, $\mathbf{J}^{(t+1)}$, $\mathbf{\Delta}_1^{(t)}$, and $\mathbf{\Delta}_2^{(t)}$, the variable $\mathbf{Z}_O^{(t+1)}$ has the following closed-form solution:

$$\begin{aligned}\mathbf{Z}_O^{(t+1)} = \big(\mathbf{I} + \mathbf{Y}_O^{\mathrm{T}}\mathbf{Y}_O\big)^{-1}\big(\mathbf{Y}_O^{\mathrm{T}}\big(\mathbf{Y}_O - \mathbf{L}_H^{(t)}\mathbf{Y}_O - \mathbf{E}^{(t)}\big) + \mathbf{J}^{(t+1)}\\ + \big(\mathbf{Y}_O^{\mathrm{T}}\mathbf{\Delta}_1^{(t)} - \mathbf{\Delta}_2^{(t)}\big)/\beta\big). \qquad (11)\end{aligned}$$

When fixing the variables $\mathbf{Z}_O^{(t+1)}$, $\mathbf{S}^{(t+1)}$, $\mathbf{E}^{(t)}$, $\mathbf{\Delta}_1^{(t)}$, and $\mathbf{\Delta}_3^{(t)}$, the $\mathbf{L}_H^{(t+1)}$ can be updated as

$$\begin{aligned}\mathbf{L}_H^{(t+1)} = \big(\big(\mathbf{Y}_O - \mathbf{Y}_O\mathbf{Z}_O^{(t+1)} - \mathbf{E}^{(t)}\big)\mathbf{Y}_O^{\mathrm{T}} + \mathbf{S}^{(t+1)}\\ + \big(\mathbf{\Delta}_1^{(t)}\mathbf{Y}_O^{\mathrm{T}} - \mathbf{\Delta}_3^{(t)}\big)/\beta\big)\big(\mathbf{I} + \mathbf{Y}_O^{\mathrm{T}}\mathbf{Y}_O\big)^{-1}. \qquad (12)\end{aligned}$$

When fixing other variables, the $\mathbf{E}^{(t+1)}$ can be updated as

$$\begin{aligned}\mathbf{E}^{(t+1)} = \frac{\lambda}{\beta}\|\mathbf{E}\|_{2,1} + \frac{1}{2}\big\|\mathbf{E} - \big(\mathbf{Y}_O - \mathbf{Y}_O\mathbf{Z}_O^{(t+1)} - \mathbf{L}_H^{(t+1)}\mathbf{Y}_O\\ + \mathbf{\Delta}_1^{(t+1)}\big)/\beta\big\|_F^2. \qquad (13)\end{aligned}$$

Three Lagrange multipliers $\mathbf{\Delta}_1^{(t+1)}$, $\mathbf{\Delta}_2^{(t+1)}$, and $\mathbf{\Delta}_3^{(t+1)}$ and the penalty parameter $\beta^{(t+1)}$ can be updated as

$$\mathbf{\Delta}_1^{(t+1)} = \mathbf{\Delta}_1^{(t)} + \beta\big(\mathbf{Y}_O - \mathbf{Y}_O\mathbf{Z}_O^{(t+1)} - \mathbf{L}_H^{(t+1)}\mathbf{Y}_O - \mathbf{E}^{(t+1)}\big) \qquad (14)$$

$$\mathbf{\Delta}_2^{(t+1)} = \mathbf{\Delta}_2^{(t)} + \beta\big(\mathbf{Z}_O^{(t+1)} - \mathbf{J}^{(t+1)}\big) \qquad (15)$$

$$\mathbf{\Delta}_3^{(t+1)} = \mathbf{\Delta}_3^{(t)} + \beta\big(\mathbf{L}_H^{(t+1)} - \mathbf{S}^{(t+1)}\big) \qquad (16)$$

$$\beta^{(t+1)} = \min(1.1\beta^{(t)}, \beta_{\max}). \qquad (17)$$

The above iterations are repeated until satisfying the convergence conditions $\|\mathbf{Y}_O - \mathbf{Y}_O\mathbf{Z}_O^{(t+1)} - \mathbf{L}_H^{(t+1)}\mathbf{Y}_O - \mathbf{E}^{(t+1)}\|_\infty \le \tau$, $\|\mathbf{Z}_O^{(t+1)} - \mathbf{J}^{(t+1)}\|_\infty \le \tau$, and $\|\mathbf{L}_H^{(t+1)} - \mathbf{S}^{(t+1)}\|_\infty \le \tau$ or the maximum iteration time $t_{\max} \ge 10^6$, where τ is the defined residual error.

The IALM for solving (7) is initialized with $\mathbf{J}^{(0)} = \mathbf{Z}_O^{(0)} = 0$, $\mathbf{L}_H^{(0)} = \mathbf{S}^{(t+1)} = 0$, $\mathbf{\Delta}_1^{(t)} = 0$, $\mathbf{\Delta}_2^{(t)} = 0$, $\mathbf{\Delta}_3^{(t)} = 0$, $\tau = 10^{-6}$, $\beta^{(0)} = 10^{-6}$, and $\beta_{max} = 10^6$. For each step in the IALM, the complexity of updating the $\mathbf{J}^{(t+1)}$ and $\mathbf{S}^{(t+1)}$ are $O(N^3)$ and $O(K^3)$ respectively; updating the variables $\mathbf{Z}_O^{(t+1)}$, $\mathbf{L}_H^{(t+1)}$, and $\mathbf{E}^{(t+1)}$ have the complexity about $O(KN^2)$ respectively; updating three Lagrange multipliers $\mathbf{\Delta}_1^{(t+1)}$, $\mathbf{\Delta}_2^{(t+1)}$, and $\mathbf{\Delta}_3^{(t+1)}$ take the complexity of $O(KN)$ respectively; and the complexity of updating the penalty parameter $\beta^{(t+1)}$ is too small and can be omitted. Therefore, the total complexity is about $O(tN^3 + tK^3 + 3tKN^2 + 3tKN)$, and it is greatly lower than the IALM on the original bands since $N < K \ll M$. After the convergence of the IALM, the coefficient matrix $\hat{\mathbf{Z}}_O = \mathbf{Z}_O^{(t+1)}$ is obtained, and the block-diagonal structure uncovers the subspaces of $\mathbf{X}$.

Algorithm 1 Procedure of Selecting a Proper Band Subset Using FLLRSC

Input: the HSI band matrix $\mathbf{X} = \{\mathbf{x}_i\}_{i=1}^N \in \mathbf{R}^{M\times N}$, the band subset size k, the projected spatial dimension K=800, the error tolerance $\tau = 10^{-6}$ and the maximum iteration time $t_{max} = 10^6$.

Step1: *Reduce the spatial dimension of $\boldsymbol{X}$ with Hadamard random projections in (3)*

Step2: *Construct the Latent low-rank representation model on $\boldsymbol{Y}_O$ in (6)*

Step3: *Solving the Latent Low-rank Representation Program (7)*

Initialization: $\mathbf{J}^{(0)} = \mathbf{Z}_O^{(0)} = 0$, $\mathbf{L}_H^{(0)} = \mathbf{S}^{(0)} = 0$, $\mathbf{\Delta}_1^{(t)} = 0$, $\mathbf{\Delta}_2^{(t)} = 0, \mathbf{\Delta}_3^{(t)} = 0$, $\beta^{(0)} = 10^{-6}$ and $\beta_{max} = 10^6$;

While ($\|\mathbf{Y}_O - \mathbf{Y}_O\mathbf{Z}_O^{(t+1)} - \mathbf{L}_H^{(t+1)}\mathbf{Y}_O - \mathbf{E}^{(t+1)}\|_\infty > \tau$, $\|\mathbf{Z}_O^{(t+1)} - \mathbf{J}^{(t+1)}\|_\infty > \tau$, $\|\mathbf{L}_H^{(t+1)} - \mathbf{S}^{(t+1)}\|_\infty > \tau$) or ($t < T$) **do**

1) Update the primal variable $\mathbf{J}^{(t+1)}$ by solving (9);
2) Update $\mathbf{S}^{(t+1)}$ by solving the optimization problem (10);
3) Update $\mathbf{Z}_O^{(t+1)}$ via the close form (11);
4) Update $\mathbf{L}_H^{(t+1)}$ via the close form (12);
5) Update $\mathbf{E}^{(t+1)}$ by solving (13);
6) Update $\mathbf{\Delta}_1^{(t+1)}$, $\mathbf{\Delta}_2^{(t+1)}$, and $\mathbf{\Delta}_3^{(t+1)}$ and $\beta^{(t+1)}$ using (14)-(17); $t \leftarrow t+1$;

End

The estimated coefficient matrix $\hat{\mathbf{Z}}_O = \mathbf{Z}_O^{(t+1)}$;

Step4: *Spectral Clustering on the Correntropy Similarity Matrix*

1) Construct the similarity matrix using the correntropy measure in (18);
2) Group the similarity matrix into k clusters using spectral clustering;
3) Select the bands whose corresponding normalized row vectors in $\mathbf{U}_k$ are closest to its cluster centroid;

Output: the band subset $\mathbf{M} = \mathbf{X}(:, \kappa)$.

D. Spectral Clustering on the Correntropy Similarity Matrix

The FLLRSC utilizes the coefficient vectors of all the observed bands to construct the similarity matrix on the graph,

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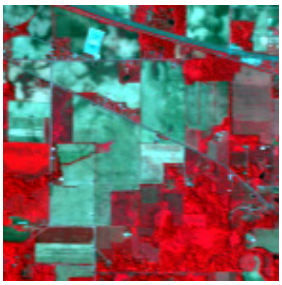


Fig. 1. Image of Indian Pines.

and then obtains the final band clusters by spectral clustering. The similarity matrix can be regarded as an undirected weighted graph $\mathbf{G} = (\mathbf{V}, \mathbf{W})$, where $v_{ij} \in \mathbf{V}$ represents the edge between pairwise bands $\mathbf{y}_O^i$ and $\mathbf{y}_O^j$, and the weight $w_{ij} \in \mathbf{W}$ measures the similarity in LRRs between them. In this article, we adopt the correntropy measure [35] to construct the similarity matrix. The correntropy is a similarity measure of signals mapped nonlinearly into a high-dimensional feature space. It takes advantage of kernel methods to compute inner products efficiently and can better identify the nonlinear characteristics of HSI bands [38]. The correntropy measure between two coefficient vectors is defined as

$$\text{Corren}_{N,\sigma}(\mathbf{z}_i, \mathbf{z}_j) = \frac{1}{K}\sum_{l=1}^{K} \Phi_\sigma(z_{il}, z_{jl}) \tag{18}$$

where $\Phi_\sigma(\cdot)$ is a kernel function. Here the Gaussian kernel $\Phi_\sigma(\mathbf{z}) = \exp(-\|\mathbf{z}\|_2^2/\sigma^2)$ is selected, and the bandwidth parameter σ of the kernel is determined as $\sigma = 0.9\delta N^{-1/5}$ [39], where δ is the minimum of the empirical standard deviation of coefficient vectors and the data interquartile range scaled by 1.34 as defined in Silverman's rule.

With the correntropy matrix $\mathbf{W} = \{\text{Corren}_{ij}\}_{i,j=1}^{N} \in \mathbb{R}^{N\times N}$, spectral clustering [40] is implemented to cluster all the projected bands on the graph into their subspaces. The symmetric normalized Laplacian matrix is built from the similarity matrix using $\mathbf{L}_{\text{sym}} = \mathbf{D}^{-1/2}\mathbf{W}\mathbf{D}^{-1/2}$, where $\mathbf{D}$ is a diagonal matrix constructed with diagonal entries of $\mathbf{W}$. The first k eigenvectors $\mathbf{U}_k = [u_1 u_2 \cdots u_k] \in \mathbb{R}^{N\times k}$ is computed through the SVD composition of the matrix $\mathbf{L}_{sym}$, and each row vector $\mathbf{h}_i$ of $\mathbf{U}_k$ is normalized into norm 1 using $h_{ij} = u_{ij}/(\sum_k u_{ik}^2)^{1/2}$. Finally, row vectors in normalized $\mathbf{U}_k$ are clustered into k clusters. The bands with row vectors closest to its cluster centroid in terms of Euclidean distance constitute the final band subset $\mathbf{M}$. The computational cost of correntropy similarity matrix is $O(N^2)$, and the computational complexity of spectral clustering is about $O(kNl)$, where l is the iteration time. Algorithm 1 summarizes the detailed procedure of FLLRSC for hyperspectral band selection.

IV. Experimental Results

A. HSI Data Description

The Indian Pines data set was taken from the Multispectral Image Data Analysis System group at Purdue University (https://engineering.purdue.edu/~biehl/MultiSpec/aviris_documentation.html). The data set was acquired by NASA on June 12, 1992 using the AVIRIS sensor from JPL. It has 20-m spatial resolutions and 10-nm spectral resolutions covering a spectrum range of 200–2400 nm. The image scene has the size 145 × 145 pixels, and covers an area of 6-mi west of West Lafayette, Indiana. After radiometric corrections and bad band removal, 200 bands were used. Fig. 1 shows sixteen classes of ground objects in the image scene.



Fig. 2. Image of PaviaU.

The PaviaU data set was taken from the Computational Intelligence Group at the Basque University (http://www.ehu.es/ccwintco/index.php/Hyperspectral_Remote_Sensing_Scenes). It was obtained from ROSIS sensor with 1.3-m spatial resolutions and 115 bands. After removing low SNR bands, 103 bands were used. A subset of a larger data set in Fig. 2 contains 610 × 340 pixels and covers the area of Pavia University, including nine classes of ground objects.

The Salinas data set was also taken from the Computational Intelligence Group at the Basque University (http://www.ehu.es/ccwintco/index.php/Hyperspectral_Remote_Sensing_Scenes). It was collected by the AVIRIS sensor over Salinas Valley, California, USA, with 3.7-m spatial resolutions and 224 bands. It was preprocessed with radiometric corrections and bad band removal, and 204 bands were used. Fig. 3 shows the image scene of size 512 × 217 pixels, which comprises of sixteen classes of ground objects.

B. Experimental Results

The standard SVM classifier and random forest (RF) are utilized on the selected bands. It adopts the radial basis function as the kernel function, and the variance parameter and penalization factor are estimated via cross-validation. The number of trees in RF is manually set to be 500. The OCA is used to quantify classification accuracies.

1) Impacts of Correntropy Matrix and HRPs in FLLRSC: To show the impacts of correntropy measure and HRPs in FLLRSC for selecting bands, three quantitative measures [25], [26] are utilized. The average information entropy (AIE) measures the information amount and evaluates the richness of spectral information in the band subset. The average

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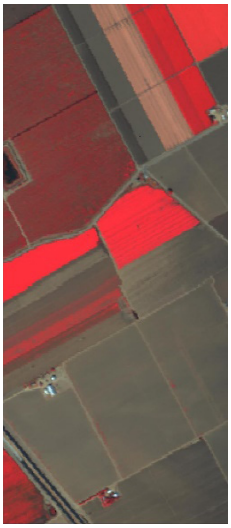


Fig. 3. Image of Salinas.

correlation coefficient (ACC) estimates the intra-band correlations. The average relative entropy (ARE) [also called average Kullback–Leibler divergence (AKLD)] is used to measure the inter-separability of selected bands and assess their distinguishability for classification. Specifically, to select k-bands, AIE, ACC, and ARE are defined as

$$\text{AIE} = \frac{1}{k}\sum_{i=1}^{k} IE(\mathbf{m}_i) \tag{19}$$

$$\text{ACC} = \frac{1}{k^2}\sum_{i=1}^{k}\sum_{j=1}^{k-1} R(\mathbf{m}_i, \mathbf{m}_j) \tag{20}$$

$$\text{ARE} = \frac{1}{k^2}\sum_{i=1}^{k}\sum_{j=1}^{k} KLD(\mathbf{m}_i \| \mathbf{m}_j) \tag{21}$$

where $IE(\mathbf{m}_i) = -\sum_{t=0}^{\max} P_t \log_2(P_t)$ is the information entropy of band vector $\mathbf{m}_i$ and P_t is the probability of value t in the histogram of $\mathbf{m}_i$,

$$R(\mathbf{m}_i, \mathbf{m}_j) = \frac{\sum_{l=1}^{M}(m_{il} - \bar{m}_i)(m_{jl} - \bar{m}_j)}{\sqrt[2]{\sum_{l=1}^{M}(m_{il} - \bar{m}_i)^2 \sum_{l=1}^{D}(m_{jl} - \bar{m}_j)^2}}$$

is the correlation coefficient between pairwise selected band vectors $\mathbf{m}_i$ and $\mathbf{m}_j$ with $\bar{m}_i$ and $\bar{m}_j$ being means of $\mathbf{m}_i$ and $\mathbf{m}_j$, respectively, $KLD(\mathbf{m}_i \| \mathbf{m}_j) = \sum_{l=1}^{M} \hat{m}_{il} \log(\hat{m}_{il}/\hat{m}_{jl})$ is the Kullback–Leibler divergence, which is the relative entropy of band $\mathbf{m}_i$ with respect to $\mathbf{m}_j$ with $\hat{m}_{il}$ being the normalized spectral response between 0 and 1 using equation $\hat{m}_{il} = (m_{il}/\sum_{l=1}^{M} m_{il})$.

The above three quantitative measures and OCAs of FLLRSC are compared with those of LLRR, "LLRR+Correntropy," and "HRP+LLRR." The difference between LLRR and FLLRSC is that the LLRR does not implement the HRP but utilizes the spectral angle measure to construct the affinity matrix. The difference between FLLRSC and "LLRR+Correntropy" is that the latter does not implement the HRP process. The "HRP+LLRR" method utilizes the spectral angle measure to construct the affinity rather than the correntropy measure. In the experiment, the number of selected bands is manually set as 30 for the three data sets. For the FLLRSC and "HRP+LLRR," the projected dimension K on the three data sets is manually set as 800. The regularization parameter λ of LLRR and FLLRSC is set as 0.1 on the Indian Pines and Salinas, and 0.05 on PaviaU.

TABLE I

COMPARISON IN QUANTITATIVE METRICS AND CLASSIFICATION ACCURACY

Datasets	Methods	Three Quantitative Metrics			SVM (%)
		AIE	ACC	ARE	OCA
Indian Pines	LLRR	9.980	0.342	28.122	79.77
	LLRR+Correntropy	10.013	0.318	31.413	82.15
	HRP+LLRR	9.865	0.326	28.124	79.70
	FLLRSC	10.134	0.303	32.077	82.31
PaviaU	LLRR	11.209	0.598	1.593	92.67
	LLRR+Correntropy	11.392	0.523	1.564	94.46
	HRP+LLRR	11.276	0.580	1.536	92.58
	FLLRSC	11.493	0.518	1.588	94.41
Salinas	LLRR	8.922	0.412	29.009	92.96
	LLRR+Correntropy	9.246	0.342	27.146	93.66
	HRP+LLRR	8.951	0.401	28.964	92.81
	FLLRSC	9.324	0.328	27.095	93.68

Table I shows that the FLLRSC and "LLRR+Correntropy" have larger AIE and ARE but slightly smaller ACC than LLRR and "HRP+LLRR." Particularly, they have about 2.5%, 2%, and 1% higher OCAs on the Indian Pines, PaviaU, and Salinas than LLRR and "HRP+LLRR" respectively. Accordingly, the FLLRSC and "LLRR+Correntropy" perform better than LLRR and "HRP+LLRR," and the Correntropy measure has significant contribution in FLLRSC for band selection. On the other hand, the FLLRSC has comparable performance with "LLRR+Correntropy," and "HRP+LLRR" shows similar results with LLRR. This indicates the dimensionality reduction via HRP affects negligibly in LLRR and FLLRSC for selecting proper bands. Therefore, the correntropy measure positively contributes to the FLLRSC while the HRP cannot.

C. Classification Performance With Different Number of Selected Bands k

This experiment investigates the classification performance of FLLRSC by changing the number of selected bands k. In the experiment, all bands and eight state-of-the-art methods are compared with the FLLRSC, including two ranking-based methods MVPCA [13] and FDPC [19], two clustering-based methods WaluDI [14] and OCF [20], two searching-based methods linear prediction (LP) [41] and orthogonal projections-based band selection (OPBS) [42], two sparsity-based methods ISSC [25] and NNLRR [32]. MVPCA, WaluDI, and LP are classical band selection methods, while ISSC, NNLRR, FDPC, OPBS, and OCF are newly proposed methods, particularly OPBS and OCF. For each method on the three data sets, the k is changed between 10 and 60 with a step interval of 5. Using cross-validation, the regularization

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TABLE II

COMPARISON IN QUANTITATIVE METRICS AMONG DIFFERENT BAND SELECTION METHODS WITH $k = 30$

Datasets	Three Quantitative Metrics	Band Selection Methods								
		MVPCA	WaluDI	LP	FDPC	ISSC	NNLRR	OCF	OPBS	FLLRSC
Indian Pines	AIE	10.005	8.742	10.188	8.993	9.703	9.085	9.451	10.649	10.134
	ACC	0.652	0.465	0.348	0.370	0.316	0.357	0.388	0.319	0.303
	ARE	13.819	31.871	24.089	31.184	32.015	20.349	29.252	16.289	32.077
	SVM (%)	70.18	77.15	73.50	77.49	81.61	70.31	75.02	71.08	82.31
PaviaU	AIE	10.871	11.048	11.233	11.253	11.209	11.114	11.277	11.017	11.493
	ACC	0.924	0.732	0.582	0.534	0.526	0.587	0.641	0.840	0.518
	ARE	0.518	1.151	1.488	1.080	1.492	1.321	1.092	0.642	1.588
	SVM (%)	86.04	92.70	93.76	91.32	94.14	92.92	88.59	86.65	94.41
Salinas	AIE	9.168	8.743	9.696	9.296	9.162	8.211	9.761	9.615	9.324
	ACC	0.865	0.535	0.356	0.352	0.406	0.674	0.416	0.435	0.328
	ARE	20.305	24.397	23.959	21.001	27.660	22.830	21.643	21.317	27.095
	SVM (%)	88.25	92.24	93.10	91.60	93.91	91.29	90.96	90.40	93.68

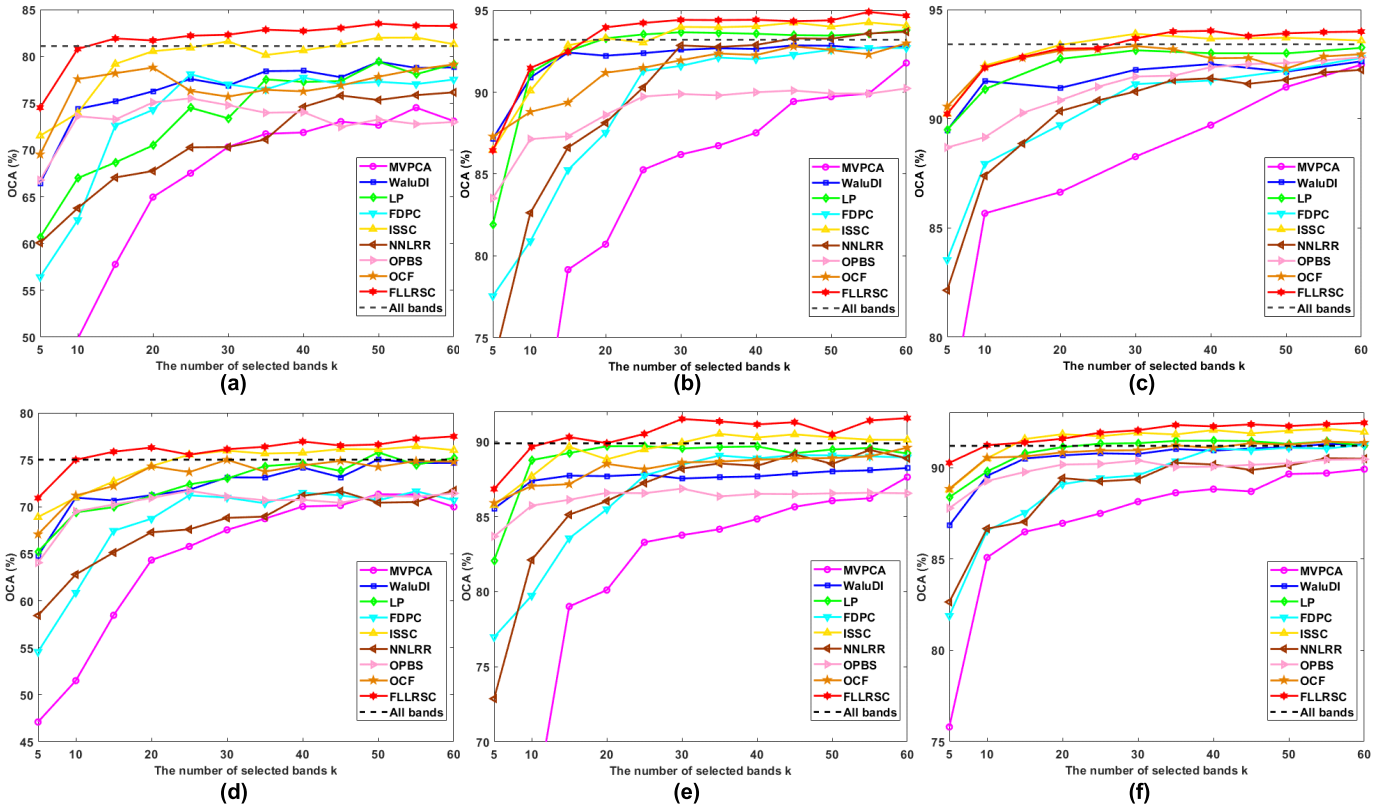


Fig. 4. OCA curves of FLLRSC, all bands and eight state-of-the-art band selection methods from SVM and RF classifiers. SVM: (a) Indian Pines, (b) PaviaU, and (c) Salinas. RF: (d) Indian Pines, (e) PaviaU, and (f) Salinas.

parameter μ of NNLRR on the Indian Pines, PaviaU, and Salinas data sets are set to be 0.01, 0.02, and 0.01, respectively. For the FLLRSC, the projected dimension K on the three data sets is manually set to be 800, and the parameter λ on the Indian Pines and Salinas is set to be 0.1, and the λ on the PaviaU is set to be 0.05. Ten percent of labeled samples from each class are randomly selected for training.

Table II compares the three quantitative metrics among different methods, where the k on all three data sets are manually set to be 30. The FLLRSC achieves higher AIE, ARE and lower ACC, and accordingly, it could select more proper bands than others. Moreover, Fig. 4 shows the OCA curves of FLLRSC, all bands, and other methods on the three data sets from SVM and RF classifiers. The FLLRSC is clearly superior to the classical NNLRR in the OCA curves of all three data sets, regardless of each classifier. The FLLRSC curve is slightly higher than that of ISSC on the Indian Pines data set, and shows similar performance with ISSC on the PaviaU and Salinas data sets. The reason for the similar performance of ISSC with FLLRSC is that the orthogonal subspace assumption of all bands and the L_2-norm of coefficient matrix guarantee that the optimal solution of coefficient matrix in ISSC is also sparse and block-diagonal, which provides correct segmentation of band subspaces and benefits selecting proper bands. The ISSC and FLLRSC curves behave better than those of the other seven methods, and the MVPCA

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TABLE III

COMPUTING TIME OF NINE BAND SELECTION METHODS WITH DIFFERENT SIZES OF BAND SUBSET k

Datasets	Number of selected Bands k	Computational times (Seconds)								
		MVPCA	WaluDI	LP	FDPC	ISSC	NNLRR	OCF	OPBS	FLLRSC
Indian Pines	k=10	0.088	596.066	23.631	2.670	0.828	32.371	5.200	0.138	36.758
	k=20	0.088	596.071	29.657	2.689	1.182	32.269	6.319	0.245	37.046
	k=30	0.089	596.076	40.021	2.703	1.563	33.180	9.478	0.332	37.530
	k=40	0.090	596.077	53.859	2.702	2.054	33.643	9.751	0.425	38.180
	k=50	0.088	596.080	71.173	2.737	5.608	34.122	10.656	0.512	38.268
PaviaU	k=10	0.298	2000.118	180.863	9.205	0.599	113.094	1.519	1.004	62.031
	k=20	0.298	2000.156	210.563	9.199	0.876	120.868	1.842	1.655	62.519
	k=30	0.308	2000.178	262.253	9.340	1.249	121.944	2.068	2.196	62.822
	k=40	0.405	2000.205	321.686	9.435	1.553	122.073	2.331	2.649	62.878
	k=50	0.440	2000.233	383.535	9.458	1.776	122.804	2.489	2.988	62.973
Salinas	k=10	0.350	3427.909	120.469	15.417	1.021	122.229	6.505	0.563	65.104
	k=20	0.362	3428.064	212.413	15.597	1.619	123.087	7.607	0.914	66.505
	k=30	0.364	3428.820	381.442	15.719	2.563	135.666	8.745	1.195	68.707
	k=40	0.382	3484.135	616.021	15.900	3.675	136.789	9.841	1.500	68.286
	k=50	0.396	3485.257	878.036	15.959	4.600	137.059	10.935	1.749	68.739

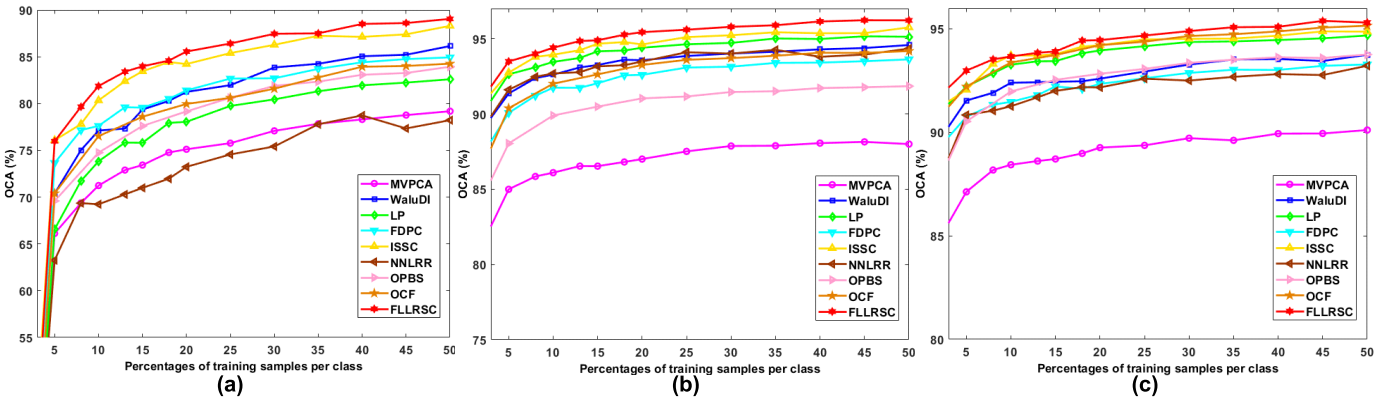


Fig. 5. Classification performance of FLLRSC and other eight band selection methods with different percentages of training samples per class. (a) Indian Pines. (b) PaviaU. (c) Salinas.

TABLE IV

OCA OF FLLRSC WITH DIFFERENT REGULARIZATION PARAMETER λ

Datasets		The regularization parameter λ										
		10^{-4}	5×10^{-4}	10^{-3}	0.005	0.01	0.05	0.1	0.5	1	5	10
Indian Pines	OCA(%)	81.51	82.16	81.85	81.59	82.29	81.60	82.31	81.76	81.98	82.43	81.93
PaviaU	OCA(%)	94.33	94.50	94.10	94.37	94.37	94.52	94.37	94.47	94.29	94.41	94.48
Salinas	OCA(%)	93.67	93.86	93.68	93.72	93.54	93.67	93.91	96.60	93.67	93.76	93.65

always performs the worst among all the OCA curves. More interestingly, the FLLRSC achieves similar or slightly higher OCAs than all bands when using a moderate band subset size (e.g., $k = 20$ on the PaviaU), whereas most other methods have lower OCAs than using all bands. The observation of FLLRSC demonstrates the advantage of band selection, that is, selecting a proper band subset could achieve similar classification accuracy of using all bands but greatly reduces the dimensionality of HSI data.

Besides, Table III lists the computational time of all nine methods, with the sizes of band subset k changing from 10 to 50 using a step interval of 10. All the methods are implemented in MATLAB 2014a and their codes are run on a WIN10 computer with Intel i5-4570 Quad Core Processor and 8 GB of RAM. The WaluDI takes the longest time, and the shortest one is MVPCA. The descending order of computational speeds of all the methods is MVPCA, OPBS, ISSC, OCF, FDPC, FLLRSC, NNLRR, LP, and WaluDI. The FLLRSC achieves the balance between computational cost and band selection performance.

D. Classification Performance of FLLRSC With Different Training Sample Sizes

This experiment compares the classification accuracies of all the above nine methods by changing the sizes of training samples. The percentages of training samples per class on the three data sets are manually changed from 2 to 50. The sizes of selected bands for all the methods are set to be 30.

As illustrated in Fig. 5, the FLLRSC and ISSC show better performance than the other seven methods, and FLLRSC

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behaves similarly with ISSC on all the three data sets. This is consistent with the observations in Section IV-B2.

E. Impacts From Regularized Parameter λ in Classification

Table IV shows the sensitivity of the FLLRSC to the regularized parameter λ, where the projected dimension K and the sizes of selected bands are 800 and 30, respectively. The OCAs of FLLRSC do not obviously change with λ. Therefore, a small λ (i.e., $\lambda < 10$) could guarantee good classification.

V. Conclusion and Future Work

This article presents the FLLRSC method to improve the performance of LRR for hyperspectral band selection. The FLLRSC assumes that all the bands are sampled from a series of latent low-rank subspaces that are spanned by both observed and unobserved bands. That can guarantee sufficiently sampling bands in representing the low-rank subspaces and improve robustness to noise. The desired bands are selected by making spectral clustering on the correntropy affinity matrix constructed from the sparse coefficient matrix. Four experiments on three hyperspectral data sets are used to testify its classification performance. Experimental results show that FLLRSC performs slightly better than ISSC and obviously better than the classical LRR (i.e., NNLRR) and other seven state-of-the-art methods with higher classification accuracy and lower computational costs. Moreover, FLLRSC achieves similar or slightly higher classification accuracy than using all bands when a moderate size of band subset is selected (e.g., $k = 20$ on the PaviaU). A small regularization parameter less than 10 can guarantee robust classification behaviors of FLLRSC. Our future work will focus on improving the computational speed of FLLRSC such that it is applicable to near-real-time on-orbit hyperspectral data processing.

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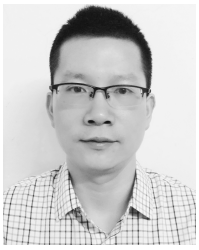
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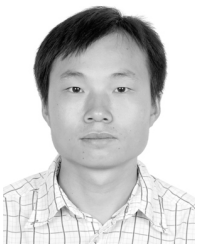
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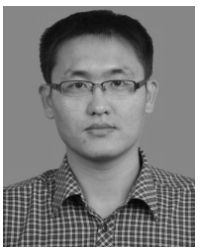
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